MAHESHWARAM ROHITH

AI/ML Engineer · LLMs, RAG & Multi-Agent Systems · Full-Stack

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SUMMARY

AI & ML undergraduate who ships production-style systems: a LoRA-tuned RAG chatbot in industry, a 6-agent autonomous QA platform, and a 3-service plagiarism-detection platform (React, Spring Boot, FastAPI). Comfortable across the stack, from model fine-tuning and computer vision to APIs, databases and deployment.

EDUCATION

Anurag University, Hyderabad 2024 – May 2028 (expected)

B.Tech, Artificial Intelligence & Machine Learning · Coursework: DSA, Machine Learning, C, Python, OOP (Java)

Sri Chaitanya Junior College, Intermediate MPC (73%), 2024 · Pratibha High School, SSC (90%), 2022

EXPERIENCE

CodeVocado — AI/ML Intern (MSME Chatbot) Oct 2025 – Jan 2026 · Remote

- Built a domain chatbot for MSMEs on **LoRA-fine-tuned Microsoft Phi-3** served via **Ollama**, grounding answers with a **RAG** pipeline to cut hallucinations on business and scheme queries.
- Added **multilingual support for 5 languages** through an automated translation pipeline, and **JWT multi-role authentication** for role-based client access.
- Optimised LLM inference for low-latency responses; earned a **Certificate of Completion and a Letter of Recommendation**.

Enigma Club, Anurag University — Coordinator, PR & Events Aug 2025 – Present

Organise technical workshops and seminars; lead PR and social-media outreach for club events.

PROJECTS

BugZero — Autonomous QA Platform | *React, Express, FastAPI, Playwright, PostgreSQL, Redis, Neo4j* GitHub

- Zero-script QA: point it at a URL and **6 specialised agents** crawl the app (BFS Playwright), click through forms and flows, and report defects with selector, HTML snippet and fix.
- Runs accessibility (axe-core, WCAG 2.1), Core Web Vitals, visual-regression and functional audits **in a single page navigation**; defects stream live over Socket.IO; Neo4j maps page relationships.
- Deterministic self-healing selectors (Levenshtein text distance + spatial decay) repair in **~15 ms at $0 LLM cost**; SSIM visual diffing, with an optional NVIDIA Eagle2-2B VLM on Hugging Face ZeroGPU.

Veritas — Document Originality Platform | *React 19, Spring Boot 3 (Java 21), FastAPI, PostgreSQL* GitHub

- Microservice platform for classroom plagiarism checks: Spring Boot manages security, classrooms, rubrics and submissions; a FastAPI engine parses PDFs, Word files and code.
- Implemented **n-gram winnowing fingerprints** and matrix similarity for 1-to-1, 1-to-many and many-to-many comparisons, with a side-by-side diff inspector in React.

WasteVision AI — Waste Classification | *TensorFlow, MobileNetV2, TFLite, OpenCV, Streamlit* GitHub

- Two-phase transfer learning on MobileNetV2 reaches **93.4% accuracy across 7 waste classes**; a 10 MB TFLite model classifies live webcam feed at **30+ FPS** with voice feedback and a statistics dashboard.

Hospital WhatsApp Automation | *React, Vite, Flask, n8n, Meta WhatsApp Business API, PostgreSQL* GitHub

- Patients book and reschedule appointments over WhatsApp via an n8n webhook → Flask state-machine bot → PostgreSQL loop, with automated 24-hour reminders; role-based dashboards for admin, reception and staff.

AI Diabetes Risk Predictor | *React (TypeScript), Flask, Supabase, scikit-learn* Live on Vercel

- Random Forest + Logistic Regression ensemble on 768 patient records (8 clinical features), **73.9% accuracy** with results in under 3 s end-to-end.

More on GitHub

- **Mind Reader AI** (Svelte, Flask): ID3 decision tree picks each question by information gain. **Kids Comic Library** (Flask, Gemini): AI-generated moral stories for children. **AI Career Advisor** (React, Firebase, Gemini 2.5, Grok): skill-gap analysis and adaptive quizzes.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL, C, HTML/CSS

AI / ML: PyTorch, TensorFlow/TFLite, scikit-learn, LLMs, RAG, LoRA fine-tuning, Ollama, Hugging Face, multi-agent systems, NLP, YOLOv8, OpenCV

Backend: FastAPI, Flask, Node.js, Express, Spring Boot, REST, Socket.IO/WebSocket, JWT auth, n8n

Frontend: React, React Native, Svelte, Vite, Tailwind CSS, Framer Motion

Data & DevOps: PostgreSQL, Redis, Neo4j, Supabase, Firebase, SQLite, Docker, Git, Playwright, Vercel

ACHIEVEMENTS & CERTIFICATIONS

- **Smart India Hackathon 2026** (Team GeoAstra): proposed SatQuery AI, an agentic vision-language assistant for satellite imagery. **SIH 2025** (Team Bit-Storm): Sarthi, an AI eco-tourism planner.
- **Avinya Hackathon**: recognised for innovation for the AI Career Advisor. **VisionNova 2.0**: national-level 24-hour hackathon.
- **Viswam AI — Summer of AI** (Meta × Swecha × IIIT Hyderabad), 4-week AI internship, 2025 · **InnoQuest 2.0** · Prompt Engineering (Simplilearn).